

Airline Customer Satisfaction Prediction Using Logistic Regression

Project Overview

This project develops a Logistic Regression model to predict airline customer satisfaction using passenger survey data. The objective is to identify the factors that most strongly influence satisfaction and provide actionable business recommendations.

1. Import Libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.metrics import f1_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay,
    roc_curve,
    roc_auc_score
)

plt.style.use('default')
pd.set_option('display.max_columns', None)
```

2. Load Dataset

```
In [2]: df = pd.read_csv('Invistico_Airline.csv')

print("Dataset Shape:", df.shape)
df.head()
```

Dataset Shape: (129880, 22)

Out[2]:

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	0	0
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	0	0
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	0	0
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	0	0
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	0	0

3. Data Understanding

In [3]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
 #   Column                                  Non-Null Count  Dtype  
---  -
 0   satisfaction                            129880 non-null object  
 1   Customer Type                          129880 non-null object  
 2   Age                                     129880 non-null int64  
 3   Type of Travel                         129880 non-null object  
 4   Class                                  129880 non-null object  
 5   Flight Distance                        129880 non-null int64  
 6   Seat comfort                           129880 non-null int64  
 7   Departure/Arrival time convenient      129880 non-null int64  
 8   Food and drink                         129880 non-null int64  
 9   Gate location                          129880 non-null int64  
10   Inflight wifi service                  129880 non-null int64  
11   Inflight entertainment                 129880 non-null int64  
12   Online support                         129880 non-null int64  
13   Ease of Online booking                 129880 non-null int64  
14   On-board service                       129880 non-null int64  
15   Leg room service                       129880 non-null int64  
16   Baggage handling                       129880 non-null int64  
17   Checkin service                       129880 non-null int64  
18   Cleanliness                           129880 non-null int64  
19   Online boarding                        129880 non-null int64  
20   Departure Delay in Minutes             129880 non-null int64  
21   Arrival Delay in Minutes               129487 non-null float64  
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB

```

In [4]: `df.describe().T`

Out[4]:

	count	mean	std	min	25%	50%	75%	max
Age	129880.0	39.427957	15.119360	7.0	27.0	40.0	51.0	85.0
Flight Distance	129880.0	1981.409055	1027.115606	50.0	1359.0	1925.0	2544.0	6951.0
Seat comfort	129880.0	2.838597	1.392983	0.0	2.0	3.0	4.0	5.0
Departure/Arrival time convenient	129880.0	2.990645	1.527224	0.0	2.0	3.0	4.0	5.0
Food and drink	129880.0	2.851994	1.443729	0.0	2.0	3.0	4.0	5.0
Gate location	129880.0	2.990422	1.305970	0.0	2.0	3.0	4.0	5.0
Inflight wifi service	129880.0	3.249130	1.318818	0.0	2.0	3.0	4.0	5.0
Inflight entertainment	129880.0	3.383477	1.346059	0.0	2.0	4.0	4.0	5.0
Online support	129880.0	3.519703	1.306511	0.0	3.0	4.0	5.0	5.0
Ease of Online booking	129880.0	3.472105	1.305560	0.0	2.0	4.0	5.0	5.0
On-board service	129880.0	3.465075	1.270836	0.0	3.0	4.0	4.0	5.0
Leg room service	129880.0	3.485902	1.292226	0.0	2.0	4.0	5.0	5.0
Baggage handling	129880.0	3.695673	1.156483	1.0	3.0	4.0	5.0	5.0
Checkin service	129880.0	3.340807	1.260582	0.0	3.0	3.0	4.0	5.0
Cleanliness	129880.0	3.705759	1.151774	0.0	3.0	4.0	5.0	5.0
Online boarding	129880.0	3.352587	1.298715	0.0	2.0	4.0	4.0	5.0
Departure Delay in Minutes	129880.0	14.713713	38.071126	0.0	0.0	0.0	12.0	1592.0
Arrival Delay in Minutes	129487.0	15.091129	38.465650	0.0	0.0	0.0	13.0	1584.0



In [5]: `df.isnull().sum()`

```
Out[5]: satisfaction      0
Customer Type           0
Age                     0
Type of Travel          0
Class                   0
Flight Distance         0
Seat comfort            0
Departure/Arrival time convenient  0
Food and drink          0
Gate location           0
Inflight wifi service   0
Inflight entertainment  0
Online support          0
Ease of Online booking  0
On-board service        0
Leg room service        0
Baggage handling        0
Checkin service         0
Cleanliness             0
Online boarding         0
Departure Delay in Minutes  0
Arrival Delay in Minutes 393
dtype: int64
```

Missing Value Treatment

The only missing values occur in 'Arrival Delay in Minutes'. We replace them with the median because delay distributions are usually skewed.

```
In [6]: df['Arrival Delay in Minutes'] = df['Arrival Delay in Minutes'].fillna(
        df['Arrival Delay in Minutes'].median()
    )

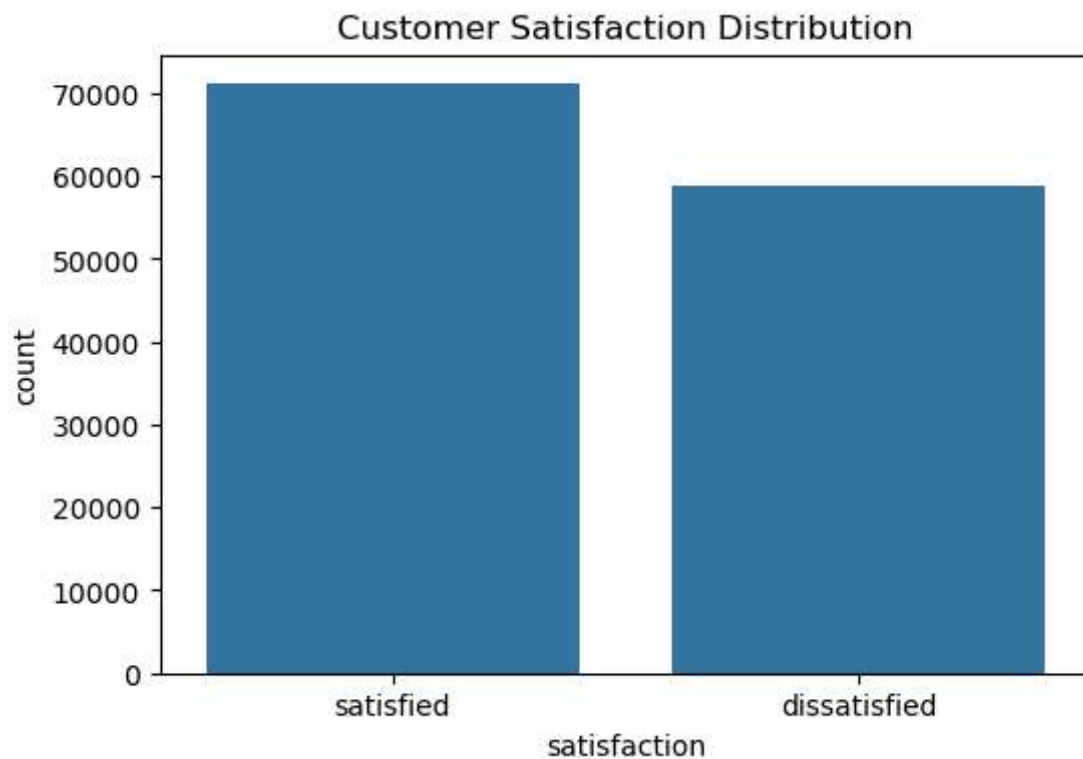
df.isnull().sum().sum()
```

```
Out[6]: np.int64(0)
```

4. Exploratory Data Analysis

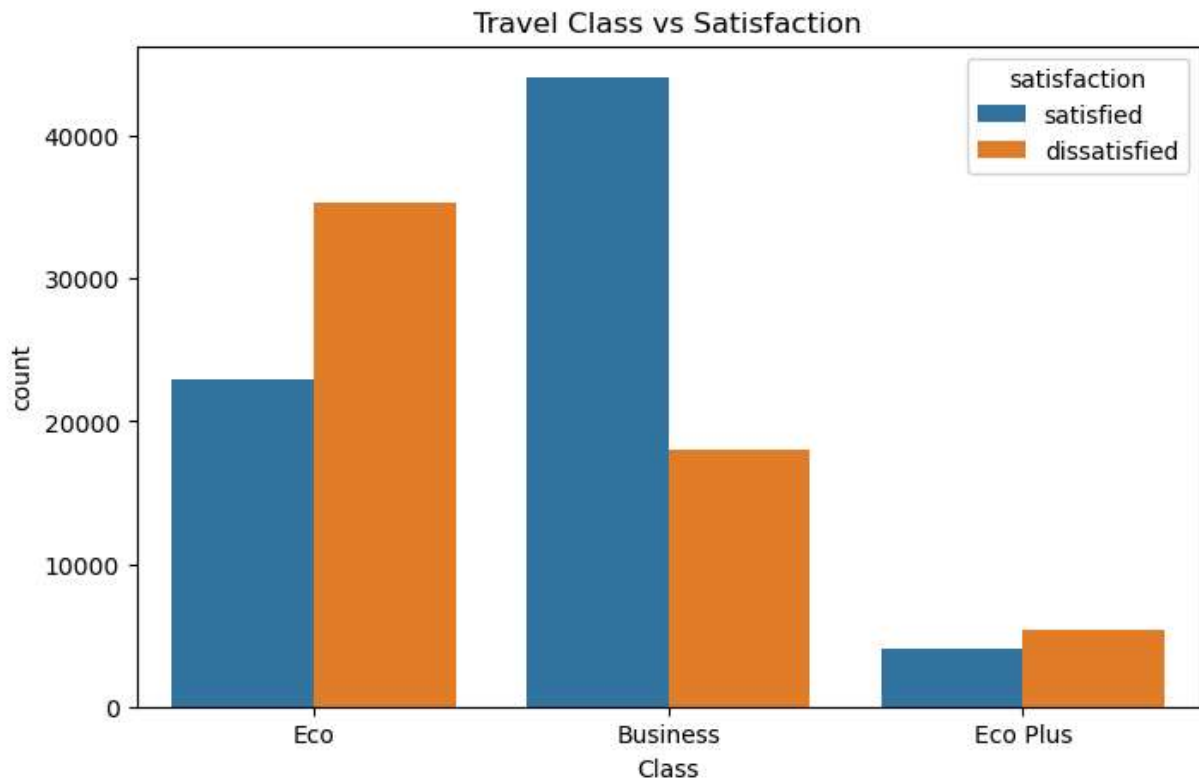
```
In [7]: plt.figure(figsize=(6,4))
sns.countplot(x='satisfaction', data=df)
plt.title('Customer Satisfaction Distribution')
plt.show()

df['satisfaction'].value_counts()
```

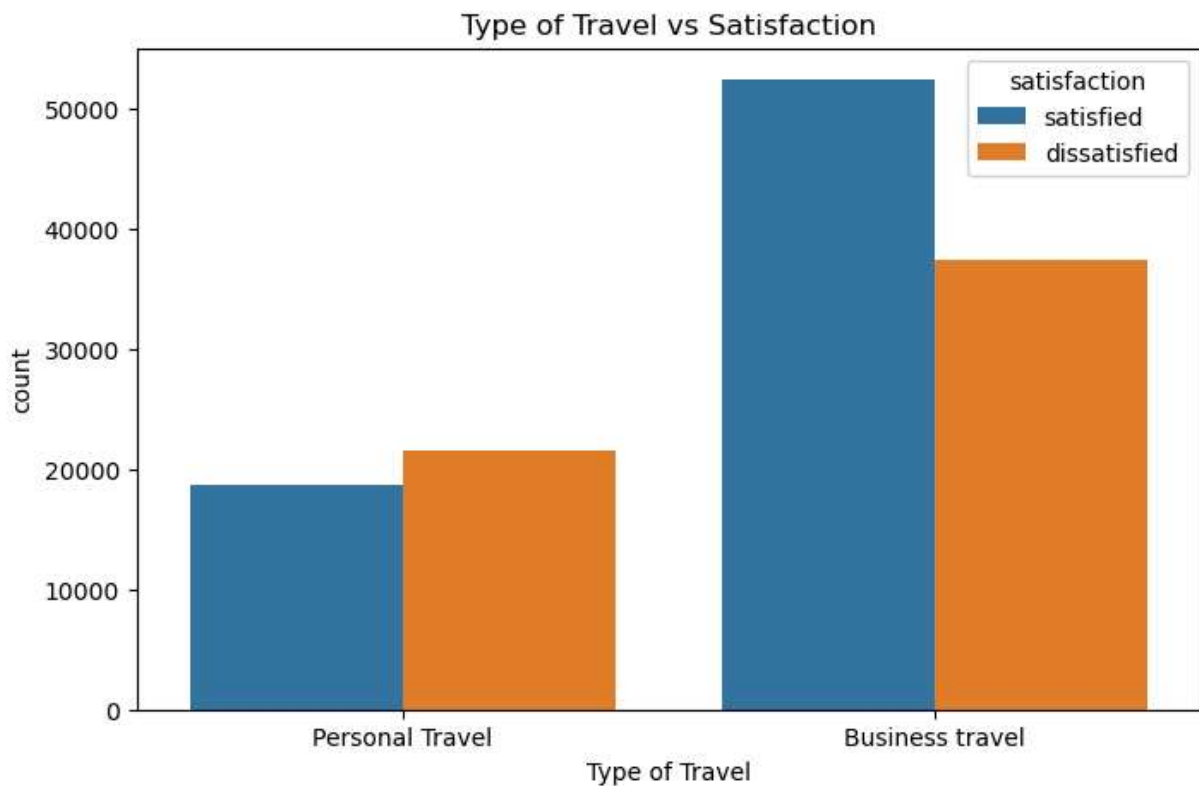


```
Out[7]: satisfaction
satisfied      71087
dissatisfied   58793
Name: count, dtype: int64
```

```
In [8]: plt.figure(figsize=(8,5))
sns.countplot(x='Class', hue='satisfaction', data=df)
plt.title('Travel Class vs Satisfaction')
plt.show()
```

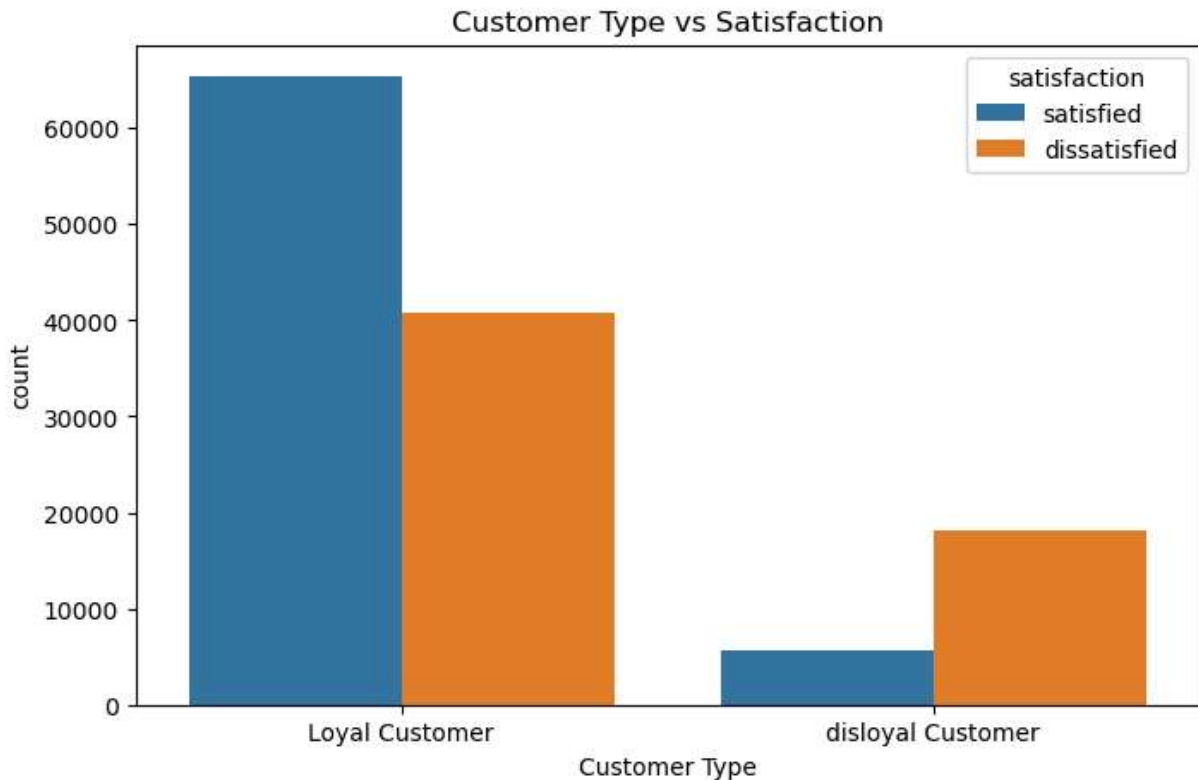


```
In [9]: plt.figure(figsize=(8,5))
sns.countplot(x='Type of Travel', hue='satisfaction', data=df)
plt.title('Type of Travel vs Satisfaction')
plt.show()
```



```
In [10]: plt.figure(figsize=(8,5))
sns.countplot(x='Customer Type', hue='satisfaction', data=df)
```

```
plt.title('Customer Type vs Satisfaction')  
plt.show()
```



5. Data Preprocessing

```
In [11]: df['satisfaction'] = df['satisfaction'].map({  
        'dissatisfied':0,  
        'satisfied':1  
    })  
  
categorical_cols = ['Customer Type', 'Type of Travel', 'Class']  
  
encoders = {}  
  
for col in categorical_cols:  
    le = LabelEncoder()  
    df[col] = le.fit_transform(df[col])  
    encoders[col] = le
```

6. Feature Selection

```
In [12]: X = df.drop('satisfaction', axis=1)  
        y = df['satisfaction']  
  
print(X.shape)
```

```
(129880, 21)
```

7. Train-Test Split

```
In [13]: X_train, X_test, y_train, y_test = train_test_split(
        X,
        y,
        test_size=0.20,
        random_state=42,
        stratify=y
    )

print(X_train.shape)
print(X_test.shape)
```

(103904, 21)

(25976, 21)

Feature Scaling

Logistic Regression performs better when numerical features are standardized.

```
In [14]: scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

8. Build Logistic Regression Model

```
In [15]: model = LogisticRegression(max_iter=2000)

model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)
y_prob = model.predict_proba(X_test_scaled)[: ,1]
```

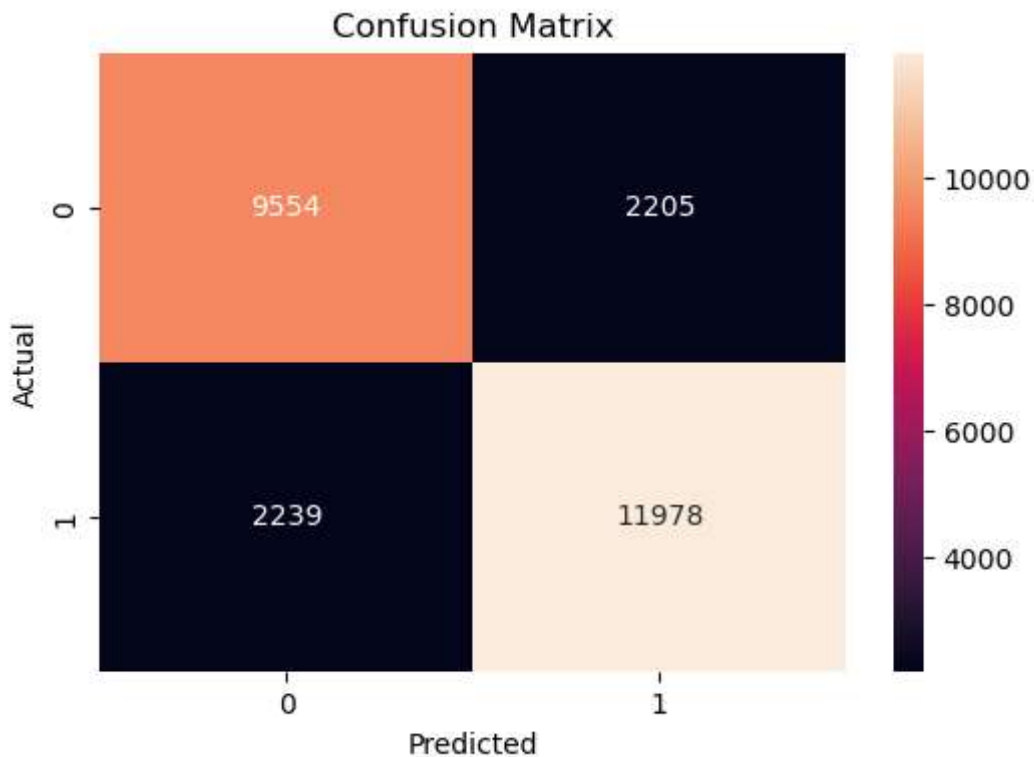
9. Model Evaluation

```
In [16]: accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(f'Accuracy : {accuracy:.4f}')
print(f'Precision: {precision:.4f}')
print(f'Recall   : {recall:.4f}')
print("F1 Score:", round(f1,4))
```


Accuracy : 0.8289
Precision: 0.8445
Recall : 0.8425
F1 Score: 0.8435

```
In [17]: cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



False Positive vs False Negative Trade-Off

The confusion matrix helps us understand the types of classification errors made by the model.

False Positive (FP)

A False Positive occurs when the model predicts that a passenger is satisfied when they are actually dissatisfied.

Business Impact:

- The airline may believe the passenger had a positive experience.

- Customer complaints may go unnoticed.
- Dissatisfied passengers may switch to competitors.
- Lost opportunities for customer recovery and retention.

False Negative (FN)

A False Negative occurs when the model predicts that a passenger is dissatisfied when they are actually satisfied.

Business Impact:

- The airline may spend resources investigating a non-existent problem.
- Additional customer support costs may be incurred.
- Less damaging than missing truly dissatisfied customers.

Which Error is More Important?

For airline customer retention, False Positives are generally more costly than False Negatives because failing to identify dissatisfied passengers can lead to customer churn, negative reviews, and reduced revenue.

Therefore, the airline should prioritize maximizing Recall for dissatisfied passengers while maintaining strong Precision.

```
In [18]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.81	0.81	0.81	11759
1	0.84	0.84	0.84	14217
accuracy			0.83	25976
macro avg	0.83	0.83	0.83	25976
weighted avg	0.83	0.83	0.83	25976

Precision Interpretation

Precision measures the percentage of passengers predicted as satisfied who were actually satisfied.

Recall Interpretation

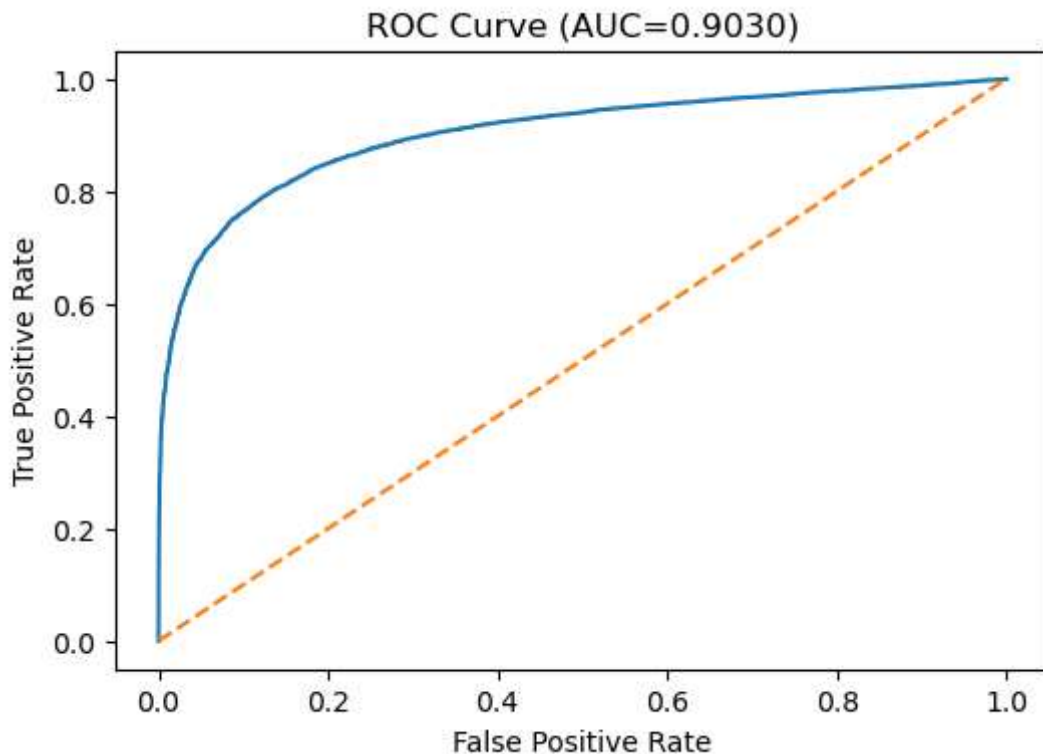
Recall measures the percentage of truly satisfied passengers correctly identified by the model.

```
In [19]: auc = roc_auc_score(y_test, y_prob)

fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(6,4))
plt.plot(fpr, tpr)
plt.plot([0,1],[0,1], '--')
plt.title(f'ROC Curve (AUC={auc:.4f})')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.show()

print("AUC:", auc)
```



AUC: 0.9029688965160622

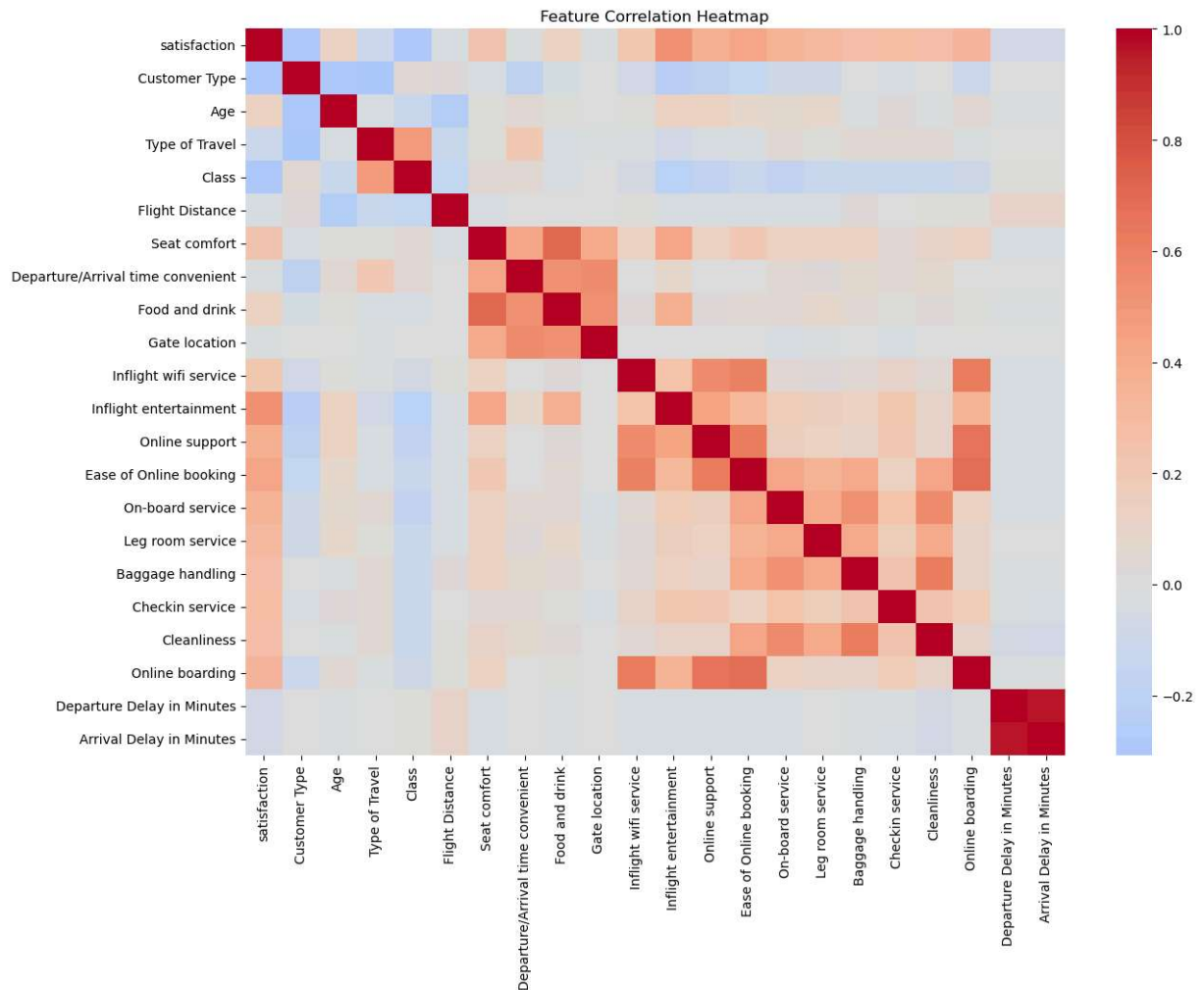
```
In [20]: # Correlation Heatmap
```

```
In [21]: plt.figure(figsize=(14,10))

corr_matrix = df.corr(numeric_only=True)

sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
    center=0
)

plt.title("Feature Correlation Heatmap")
plt.show()
```



10. Coefficient Interpretation

```
In [22]: coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_[0]
})

coefficients = coefficients.sort_values(
    by='Coefficient',
    ascending=False
)

coefficients.head(15)
```

Out[22]:

	Feature	Coefficient
10	Inflight entertainment	0.980771
13	On-board service	0.393420
5	Seat comfort	0.382090
16	Checkin service	0.359930
12	Ease of Online booking	0.323253
14	Leg room service	0.310017
18	Online boarding	0.181789
8	Gate location	0.161373
11	Online support	0.151010
15	Baggage handling	0.114572
19	Departure Delay in Minutes	0.074304
17	Cleanliness	0.071090
9	Inflight wifi service	-0.125191
1	Age	-0.129818
4	Flight Distance	-0.172818

Top Positive and Negative Drivers

```
In [23]: print("Top Positive Drivers")
display(coefficients.head(10))

print("Top Negative Drivers")
display(coefficients.tail(10))
```

Top Positive Drivers

	Feature	Coefficient
10	Inflight entertainment	0.980771
13	On-board service	0.393420
5	Seat comfort	0.382090
16	Checkin service	0.359930
12	Ease of Online booking	0.323253
14	Leg room service	0.310017
18	Online boarding	0.181789
8	Gate location	0.161373
11	Online support	0.151010
15	Baggage handling	0.114572

Top Negative Drivers

	Feature	Coefficient
17	Cleanliness	0.071090
9	Inflight wifi service	-0.125191
1	Age	-0.129818
4	Flight Distance	-0.172818
20	Arrival Delay in Minutes	-0.265061
7	Food and drink	-0.294136
3	Class	-0.294970
6	Departure/Arrival time convenient	-0.325437
2	Type of Travel	-0.414641
0	Customer Type	-0.751735

Interpretation of Coefficients

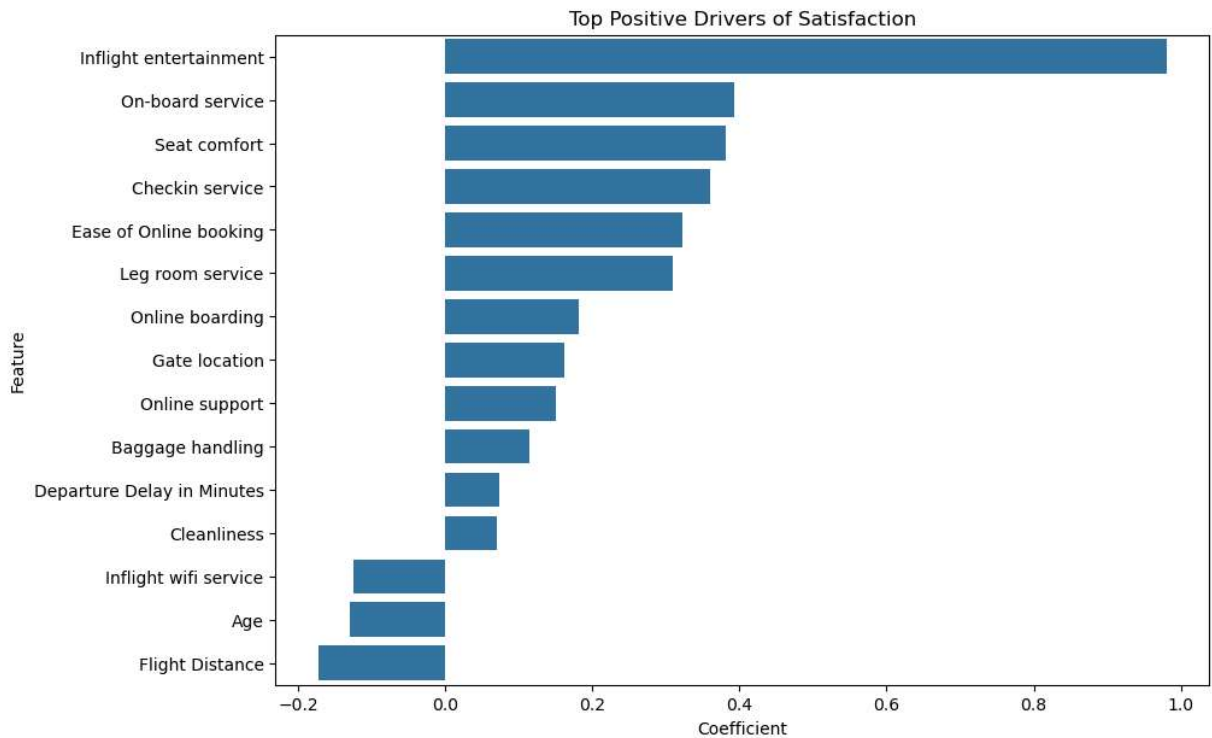
Positive coefficients increase the likelihood of passenger satisfaction.

Negative coefficients decrease the likelihood of passenger satisfaction.

Features such as Online Boarding, Inflight Wi-Fi Service, Online Support, Seat Comfort, and Inflight Entertainment are expected to be among the strongest positive drivers.

Features associated with delays and service inconvenience tend to decrease satisfaction.

```
In [24]: plt.figure(figsize=(10,7))
sns.barplot(
    data=coefficients.head(15),
    x='Coefficient',
    y='Feature'
)
plt.title('Top Positive Drivers of Satisfaction')
plt.show()
```



```
In [25]: coefficients.tail(15)
```

Out[25]:

	Feature	Coefficient
18	Online boarding	0.181789
8	Gate location	0.161373
11	Online support	0.151010
15	Baggage handling	0.114572
19	Departure Delay in Minutes	0.074304
17	Cleanliness	0.071090
9	Inflight wifi service	-0.125191
1	Age	-0.129818
4	Flight Distance	-0.172818
20	Arrival Delay in Minutes	-0.265061
7	Food and drink	-0.294136
3	Class	-0.294970
6	Departure/Arrival time convenient	-0.325437
2	Type of Travel	-0.414641
0	Customer Type	-0.751735

Odds Ratio Interpretation

```
In [26]: odds_ratios = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_[0]
})

odds_ratios["Odds Ratio"] = np.exp(
    odds_ratios["Coefficient"]
)

odds_ratios.sort_values(
    by="Odds Ratio",
    ascending=False
).head(15)
```


Out[26]:

	Feature	Coefficient	Odds Ratio
10	Inflight entertainment	0.980771	2.666512
13	On-board service	0.393420	1.482040
5	Seat comfort	0.382090	1.465345
16	Checkin service	0.359930	1.433229
12	Ease of Online booking	0.323253	1.381615
14	Leg room service	0.310017	1.363448
18	Online boarding	0.181789	1.199361
8	Gate location	0.161373	1.175124
11	Online support	0.151010	1.163008
15	Baggage handling	0.114572	1.121394
19	Departure Delay in Minutes	0.074304	1.077134
17	Cleanliness	0.071090	1.073678
9	Inflight wifi service	-0.125191	0.882329
1	Age	-0.129818	0.878256
4	Flight Distance	-0.172818	0.841291

Odds Ratio Interpretation

Odds ratios provide a more intuitive interpretation of Logistic Regression coefficients.

- Odds Ratio > 1 indicates increased likelihood of satisfaction.
- Odds Ratio < 1 indicates decreased likelihood of satisfaction.

For example, if a feature has an odds ratio of 1.5, a one-unit increase in that feature increases the odds of customer satisfaction by approximately 50%.

11. Business Recommendations

Recommendation 1

Improve inflight Wi-Fi quality and reliability.

Recommendation 2

Enhance online support and booking systems.

Recommendation 3

Improve seat comfort and onboard experience.

Recommendation 4

Reduce departure and arrival delays.

Recommendation 5

Focus on customer experience initiatives for frequent and business travelers.

12. Limitations

1. Logistic Regression assumes a linear relationship in the log-odds.
2. Survey responses may contain subjective bias.
3. Correlated service-rating variables may affect coefficient interpretation.
4. Future customer preferences may differ from historical data.

13. Future Work

1. Random Forest Classifier
2. Gradient Boosting
3. XGBoost
4. Hyperparameter Tuning
5. Customer Segmentation
6. Real-Time Prediction Dashboard

14 Feature Engineering Discussion

Feature Engineering

Several preprocessing steps were performed before model development:

1. Missing values in Arrival Delay in Minutes were replaced using the median value.
2. Categorical variables were encoded using Label Encoding.
3. The target variable (satisfaction) was converted into binary form:
 - Satisfied = 1
 - Dissatisfied = 0
4. Features were standardized using StandardScaler to improve Logistic Regression performance.

5. The dataset was split into training and testing sets using stratified sampling to preserve class distribution.

Model Assumption Discussion

Logistic Regression Assumptions

Logistic Regression relies on several assumptions:

1. Binary target variable.
2. Independent observations.
3. Limited multicollinearity among predictors.
4. Linear relationship between predictors and the log-odds of the outcome.
5. Large sample size.

In this project:

- The target variable is binary (Satisfied/Dissatisfied).
- The dataset contains over 129,000 observations.
- Feature scaling was performed to improve optimization.
- Future work should include Variance Inflation Factor (VIF) analysis to investigate multicollinearity.

15. Summary

Executive Summary

This project developed a Logistic Regression model to predict airline customer satisfaction using passenger survey data. After preprocessing, encoding categorical variables, and splitting the data into training and testing sets, a Logistic Regression classifier was trained and evaluated using Accuracy, Precision, Recall, Confusion Matrix, and ROC-AUC metrics.

The analysis identified service-related factors such as Inflight Wi-Fi Service, Online Boarding, Online Support, Seat Comfort, and Inflight Entertainment as major drivers of customer satisfaction. The results provide actionable insights for airline management to improve customer experience and increase passenger retention.